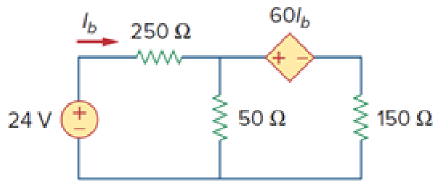
Chapter 3, Problem 9P

Problem

Determine *Ib* in the circuit in Fig. 3.58 using nodal analysis.

Figure 3.58:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.58 in the textbook.

Draw the circuit with node notations.

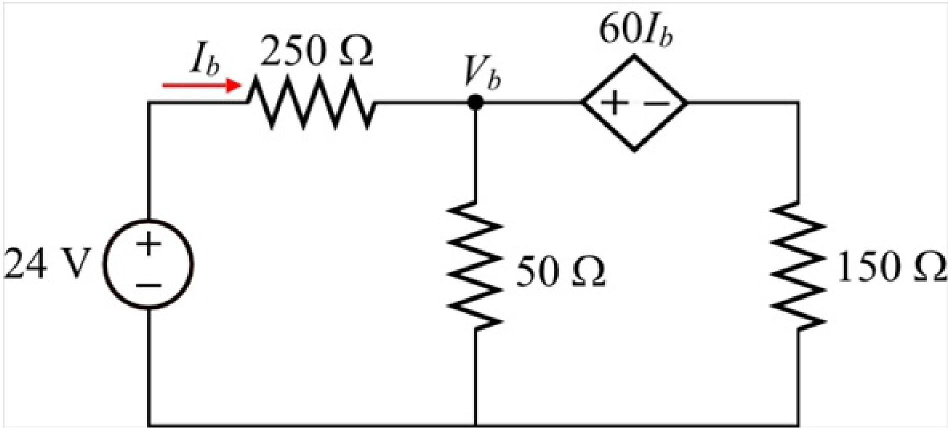


Figure 1

Step 2 of 4

Write the Kirchhoff's current law equation at node, V_b.

$$\frac{V_b - 24}{250} + \frac{V_b}{50} + \frac{V_b - 60I_b}{150} = 0$$

$$\frac{3V_b - 72 + 15V_b + 5V_b - 300I_b}{750} = 0$$

$$23V_b - 300I_b = 72 \quad \ldots\ldots \text{(1)}$$

Step 3 of 4

From Figure 1, the current through $250\,\Omega$ resistor is,

$$I_b = \frac{24 - V_b}{250}$$

$$250I_b = 24 - V_b$$

$$V_b = 24 - 250I_b \quad \ldots\ldots \text{(2)}$$

Substitute $24-250I_b$ for V_b in equation (1).

$$23(24-250I_b)-300I_b=72$$
$$552-5750I_b-300I_b=72$$
$$(-5750-300)I_b=72-552$$
$$-6050I_b=-480$$

$$\begin{aligned} I_b &= \frac{-480}{-6050} \\ &= 0.07934\text{ A} \\ &= 79.34\text{ mA} \end{aligned}$$

Therefore, the value of current, I_b is $\boxed{79.34\text{ mA}}$.